Roll No. ....

5th Sem / Ceramic

Subject : Modern Ceramics

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 _______ are used to absorb fission neutrons in nuclear reactors

a) Controls Rods b) Carbon

c) Graphite d) All of the above

Q.2 Modern ceramics are made from

a) Pure Material b) Normal material

c) Impure material d) None of these

Q.3 Superconductivity was invented by

a) H.K Onnes b) Delta

c) Rutherford d) Dalton

Q.4 ______ are used to reduce speed of fast fission neutrons

a) Moderators b) Coolent

c) Control Rods d) All of the above

Q.5 Classification of modern ceramics include______

a) Superconductors b) Nuclear Ceramics

c) Bio Ceramics d) All of these

Q.6 Sensors are

a) Resistors b) Variastors

c) Thermistors d) All of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Super conductors have_____resistance(Zero/High)

Q.8 Bio-ceramics are used for repair and reconstruction of damaged body parts.(True/False)

Q.9 PLZT stands for phosphorous, lanthanum, zirconate and titanate.(True/False)

Q.10 Hand ferrites are used in____.(Speakers, Inductors)

Q.11 Modern ceramics are used in nuclear reactors. (True/False)

Q.12 Uranium dioxide is used as fuel in nuclear reactor. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain scope of newer ceramics.

Q.14 List applications of super conductors.

Q.15 Give classification of modern ceramics.

Q.16 Explain meissner effect.

Q.17 Write functions of control rods.

Q.18 Compare organic and in organic membrane.

Q.19 Write uses of grinding wheels.

Q.20 Applications of dental ceramics.

Q.21 Write application of Hard ferrites

Q.22 Classify Bio Ceramics. Explain any one.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain manufacture of Hard Ferrites.

Q.24 Write different ceramics materials used in nuclear reactor.

Q.25 How will you manufacture thermistors?